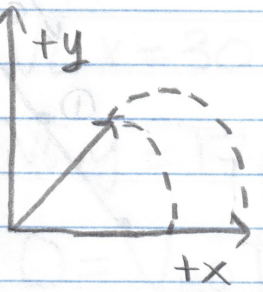
PHYS 2500 Lecture 21: "WorkSheet 7"

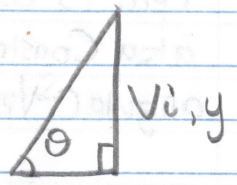
P. 226-233
problem 1

Known: mass = 250 g = 0.25 kg
$\theta = 30°$, $V_i = 35$ m/s

Looking at the drag force and its magnitude



$V_{i,x} = V_i \cos\theta$
$V_{i,y} = V_i \sin\theta$



w/o drag vs. w/ drag

$\sin\theta = \frac{V_{i,y}}{V_i}$

Constant acceleration?

$-F_{net,x} = m \cdot a_x = -F_d$ ($a_x = \frac{-F_{net,x}}{m}$)

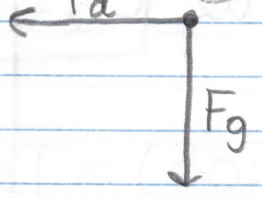
does $F_{net,y}$ cancel?

$F_{net,y} = m \cdot a_y = -g$

$F_{net,y} = -mg$

$a_y = -g$

FBD →



↳ "Constant horizontal drag force"

$V_{i,x} = 35\cos(30^\circ)$

$V_{i,y} = 35\sin(30^\circ)$

$V_{i,x} = 30.3\ m/s$

$V_{i,y} = 17.5\ m/s$

Use y-motion to find flight time

$0 = V_{y0}T - \frac{1}{2}gT^2$

$T = \frac{2V_{y0}}{g}$

$= \frac{2(17.5\ m/s)}{(9.8\ m/s)}$

$= \frac{35\ m/s}{9.8\ m/s^2}$

$= 3.57\ s$

Find range without wind (baseline)

$R_0 = V_{x0}T = (30.3\ m/s)(3.57\ s)$

$= 108.2\ m$

$\frac{m}{s} \cdot \frac{s}{}$

Apply the 20% Condition

$\frac{1}{2}aT^2 = 0.2(108.2\,m)$

$\approx 21.65\,m$

$a = \frac{2(21.65\,m)}{(3.57\,s)^2}$

$= 3.40\,m/s^2$

$F = ma$

$kg\,m/s^2 = N$

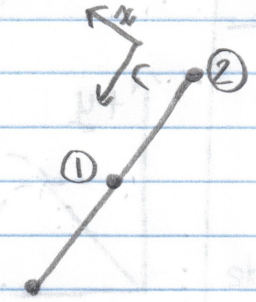
$F = ma$

$= (0.250\,kg)(3.40\,m/s^2)$

$= 0.85\,N$

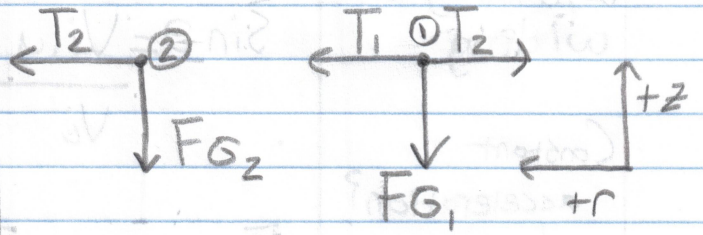
p. 227
problem 2

$\omega_{RPM} = 30\,RPM$



Uniform circular motion if rpm is constant aka Constant angular Velocity

r t z Coordinates



Only care about tension forces in this question.

$F_{net,r,2} = m_2 \omega^2 R_2$

$R_2 = 2.0\,m$

$a_r = \omega^2 R$

$F_{net,r,1} =$